

A TWO-LEVEL DIGIT LAW FOR APÉRY'S PAIR

[AUTHOR]

ABSTRACT. Let a_n be the Apéry numbers for $\zeta(3)$ and b_n their companion. We show that the pair $(a_n, p^3 b_n)$ transforms under base- p digit splitting by a single upper-triangular matrix whose diagonal is one explicit scalar and whose off-diagonal entry is the low digit's own companion value, scaled by exactly p^3 . The scalar is the second-order truncation of a Γ -deformation of the summand, summed over the full digit range. We prove the diagonal of the law to second order, for both rows, and with it the rank-one profile of the digit defect. Everything reduces to the residue theorem for one explicit rational function, applied twice: once over $\mathbb{C}$, giving an identity over $\mathbb{Q}$, and once over $\mathbb{F}_p$, where the carry region is exactly the complement of the pole divisor. Only the order- p^4 off-diagonal remains conjectural.

Throughout p is a prime ≥ 5 , and for $x, y \in \mathbb{Q}$ we write $x \equiv y \pmod{p^m}$ to mean $v_p(x - y) \geq m$. Evidence is labelled: statements called Theorem, Lemma, Proposition or Corollary are proved; Conjectures are not, and the single Conjecture below records exactly what is not.

1. THE PAIR, AND THE TWO FUNCTIONALS

Definition 1.1. $A(n, k) = \binom{n}{k}^2 \binom{n+k}{k}^2$, $a_n = \sum_{k=0}^n A(n, k)$, $H_m^{(r)} = \sum_{j=1}^m j^{-r}$, $H_m = H_m^{(1)}$, and $b_n = \sum_{k=0}^n A(n, k)(2H_n^{(3)} - H_k^{(3)})$, the solution of

$$(n+2)^3 b_{n+2} = (34n^3 + 153n^2 + 231n + 117) b_{n+1} - (n+1)^3 b_n, \quad b_0 = 0, \quad b_1 = 6.$$

Definition 1.2. $u(r, s) = H_{r+s} - H_{r-s}$ and $v(r, s) = H_{r+s} + H_{r-s} - 2H_s$; these are $\frac{1}{2}\partial_n \log A$ and $\frac{1}{2}\partial_k \log A$. Put

$$U_r = \sum_s A(r, s) u(r, s), \quad V_r = \sum_s A(r, s) v(r, s).$$

$U_0, \dots, U_8 = 0, 6, 105, 2219, \frac{104825}{2}, \frac{13276637}{10}, \frac{70543291}{2}, \frac{67890874657}{70}, \frac{766399019471}{28}$. Note $U_r = \frac{1}{2} da_n/dn$ in the Γ -form.

2. THE EXPANSION, AND WHY THE DEFECT IS RANK ONE

Lemma 2.1 (digit expansion). *For $0 \leq s \leq r < p$, $0 \leq c \leq a < p$ and $r + s < p$,*

$$A(ap + r, cp + s) \equiv A(a, c) A(r, s) (1 + 2p[a u(r, s) + c v(r, s)]) \pmod{p^2}.$$

Outside that region the relevant binomial is $\equiv 0 \pmod{p}$; since A is a square, both $A(ap + r, cp + s)$ and $A(r, s)$ are then $\equiv 0 \pmod{p^2}$, so those regions do not contribute at first order at all.

Proof. Only the weak Wolstenholme congruence $H_{p-1} \equiv 0 \pmod{p}$ is needed (the $(ip)^2$ terms die of their own accord), and it follows from pairing $l \leftrightarrow p - l$. It makes the $(ap)!/(p^a a!) \equiv ((p-1)!)^a$ factors cancel, giving $\binom{ap+r}{cp+s} \equiv \binom{a}{c} \binom{r}{s} (1 + p[aH_r - cH_s - (a-c)H_{r-s}])$ and $\binom{(a+c)p+(r+s)}{cp+s} \equiv \binom{a+c}{c} \binom{r+s}{s} (1 + p[(a+c)H_{r+s} - cH_s - aH_r])$ modulo p^2 . In the product A the two aH_r terms cancel. $\square$

Lemma 2.1 is available only on $r + s < p$, so summing it produces the *restricted* functionals $\tilde{U}_r = \sum_{s \leq r, r+s < p} A(r, s) u(r, s)$ and $\tilde{V}_r$, not U_r and V_r . The two agree:

Lemma 2.2. For $r < p$, $\tilde{U}_r \equiv U_r$ and $\tilde{V}_r \equiv V_r \pmod{p}$.

Proof. For $r+s \geq p$ the carried binomial satisfies $\binom{r+s}{s} \equiv 0 \pmod{p}$, and A is a square, so $p^2 \mid A(r, s)$; while $u(r, s), v(r, s)$ have arguments $< 2p$ and hence $v_p \geq -1$. Every dropped term therefore has $v_p \geq 1$. $\square$

This is not a formality: U_r is a p -independent rational which *does* carry p in its denominator for small r — $U_5 = 13276637/10$ and $U_7 = 67890874657/70$, so $v_5(U_5) = v_7(U_7) = -1$. It is p -integral precisely because $r < p$, which is a consequence of Lemma 2.2 rather than of anything stated before it.

Since $H_{ap+r}^{(3)} = p^{-3}H_a^{(3)} + G_{ap+r}$ with $G_m = \sum_{j \leq m, p \nmid j} j^{-3}$ p -integral, the weight contributes nothing below order p^3 . Lemma 2.1 therefore gives, with $a'_a = \sum_c c A(a, c)$ and $b'_a = \sum_c c A(a, c)(2H_a^{(3)} - H_c^{(3)})$,

$$\frac{1}{p}(a_{ap+r} - a_a a_r) \equiv 2[a a_a U_r + a'_a V_r], \quad \frac{1}{p}(p^3 b_{ap+r} - b_a a_r) \equiv 2[a b_a U_r + b'_a V_r] \pmod{p},$$

a form of rank at most two. One channel degenerates identically:

Theorem 2.3. $V_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (H_{n+k} + H_{n-k} - 2H_k) = 0$ for all $n \geq 0$.

Proof. $V_n = \frac{1}{2} \sum_k \partial_k A_\Gamma(n, k)$ with $A_\Gamma(n, z) = [\Gamma(n+z+1)/(\Gamma(z+1)^2 \Gamma(n-z+1))]^2$, and reflection gives $A_\Gamma(n, z) = (\sin^2 \pi z / \pi^2) g(z)$ with $g(z) = \Gamma(n+z+1)^2 \Gamma(z-n)^2 / \Gamma(z+1)^4$. No analysis is required, because g is *rational*:

$$g = \varphi^2, \quad \varphi(z) = \frac{\prod_{i=1}^n (z+i)}{\prod_{j=0}^n (z-j)}.$$

At an integer $z = k$ the double poles of A_Γ cancel and $\partial_z A_\Gamma|_{z=k} = \text{Res}_{z=k} g$. Now φ has simple poles with residues $\beta_k = (-1)^k \binom{n}{k} \binom{n+k}{k}$ — this is Lagrange interpolation of $\prod_{i=1}^n (X+i)$ at the nodes $0, \dots, n$ — so

$$\text{Res}_{z=m} \varphi^2 = 2\beta_m \sum_{k \neq m} \frac{\beta_k}{m-k},$$

which is antisymmetric in (m, k) and therefore sums to zero over m . $\square$

Remark 2.4. No telescoping proof of Theorem 2.3 of the shape used for the closed form exists: any $\Psi = A \cdot (\alpha H_{n+k} + \beta H_{n-k} + \gamma H_k + \delta)$ forces $r(k)\alpha(k+1) - \alpha(k) = 1$, which is Gosper's equation for the Apéry summand.

Corollary 2.5. For $a, r < p$,

$$\frac{1}{p}(a_{ap+r} - a_a a_r) \equiv 2a a_a U_r, \quad \frac{1}{p}(p^3 b_{ap+r} - b_a a_r) \equiv 2a b_a U_r \pmod{p}.$$

Both rows carry the same functional U_r and differ only by their own value at level a . Equivalently

$$(a_n, p^3 b_n) \equiv (a_a, b_a) \cdot (a_r + 2p a U_r) \pmod{p^2}, \quad n = ap + r.$$

Remark 2.6. U_r is not a_r , nor b_r , nor divisible by a_r : at $p = 5$, $U_r \neq 0$ exactly where $a_r \equiv 0$. It is the exact analogue, for this pair, of the functional appearing in the corresponding law for the Brown–Zudilin $\zeta(5)$ family, and has the same origin, $\partial_n \log$ of the summand.

3. THE SCALAR, AND THE MATRIX

Definition 3.1. With $\Pi(t) = \prod_{i \leq t} (1 + \varepsilon/i)$ and $\bar{\Pi}(t) = \prod_{i \leq t} (1 - \varepsilon/i)$,

$$A_\Gamma(r + \varepsilon, s) = A(r, s) \frac{\Pi(r + s)^2}{\Pi(r - s)^2} \quad (s \leq r),$$

$$A_\Gamma(r + \varepsilon, r + m) = \varepsilon^2 C(r, m) \Pi(2r + m)^2 \bar{\Pi}(m - 1)^2 \quad (m \geq 1),$$

where $C(r, m) := ((2r + m)! (m - 1)! / ((r + m)!)^2)^2$, and $\text{Adig}(p, r; \varepsilon) = \sum_{s=0}^{p-1} A_\Gamma(r + \varepsilon, s)$, the sum being over the *full* digit range. Write $X_p(r) = [\varepsilon^2] \text{Adig}(p, r)$.

Then $[\varepsilon^0] = a_r$ and $[\varepsilon^1] = 2U_r$. The terms with digit $s > r$ — the borrow region — are present and are exactly of order ε^2 .

Theorem 3.2 (the two-level digit law). *Let $p \geq 5$ be prime and $n = ap + r$ with $0 \leq a, r < p$. With $u(a, r) = a_r + 2pa U_r + p^2 a^2 X_p(r)$,*

$$(a_n, p^3 b_n) \equiv (a_a, b_a) \cdot u(a, r) \pmod{p^3};$$

equivalently $u(a, r) = [\sum_{s=0}^{p-1} A_\Gamma(r + \varepsilon, s)]_{\varepsilon=pa}$ truncated at order ε^2 . The floor is exactly 3.

The proof is §4. Truncating Adig at orders $m = 0, 1, 2$ gives floors 1, 2, 3, and floor 3 for every $m \geq 3$: the scalar picture saturates at depth three. The restricted deformation $\sum_{s \leq r}$ saturates at depth two, so the borrow region is exactly the second-order correction.

Corollary 3.3. *The digit defect has rank ≤ 1 at first order and rank ≤ 1 at second order, for both rows, with a -side factor exactly $a \cdot (a_a \mid b_a)$ respectively $a^2 \cdot (a_a \mid b_a)$, and with the same r -side vector for both rows: $(U_r)_r$ at first order, $(X_p(r))_r$ at second. The rank is exactly 1 iff the respective r -vector is $\neq 0$, which holds at all 13 primes $5 \leq p \leq 47$.*

Proposition 3.4 (the cross term). *Modulo p a full block satisfies $\sum_{t=1}^{p-1} t^{-3} \equiv 0$, so $G_{ap+r} \equiv H_r^{(3)}$ and $G_{cp+s} \equiv H_s^{(3)}$. Hence the piece of $p^3 b_n$ neglected above is*

$$p^3 \sum_k A(n, k) (2G_n - G_k) \equiv p^3 a_a b_r \pmod{p^4}.$$

That is, the low digit acts on the pair by

Conjecture 3.5. For every prime $p \geq 5$ and $n = ap + r$ with $a, r < p$,

$$(a_n, p^3 b_n) \equiv (a_a, b_a) \cdot \begin{pmatrix} u(a, r) & p^3 b_r \\ 0 & u(a, r) \end{pmatrix} \pmod{p^4},$$

with $u(a, r)$ as in Theorem 3.2.

Its diagonal is now proved: the mod p^2 truncation is Corollary 2.5 and the mod p^3 truncation is Theorem 3.2. What remains conjectural is only the off-diagonal entry at order p^4 . That entry is derived in Proposition 3.4 and verified at 13 primes in the sharp form: at order p^3 the b -row defect has rank 3, subtracting $p^3 a_a b_r$ drops it to rank 2, and its r -side row space then coincides with the a -row's. The residual scalar defect at p^3 has rank 2 and is not yet named; that alone is why the statement is still a conjecture.

Remark 3.6. The cross entry is the low digit's own companion value, scaled by exactly $p^3 = p^w$ where $w = 3$ is the weight of $\zeta(3)$. So the weight is precisely the order at which the off-diagonal switches on. For the rank-three Brown–Zudilin $\zeta(5)$ family the corresponding statement is that a cross term cannot appear before order p^3 ; here it is derived rather than bounded, and the entry is named.

4. THE SECOND ORDER

Writing $A(ap + r, cp + s) \equiv A(a, c)A(r, s) \exp(2p\Lambda_1 - p^2\Lambda_2)$ with $\Lambda_1 = au + cv$ and $\Lambda_2 = (a + c)^2 H_{r+s}^{(2)} - 2c^2 H_s^{(2)} - (a - c)^2 H_{r-s}^{(2)}$, the bracket $2\Lambda_1^2 - \Lambda_2$ has three channels, and we write $\Sigma_{a^2}, \Sigma_{ac}, \Sigma_{c^2}$ for their $A(r, s)$ -weighted sums over $s \leq r$:

$$\begin{aligned} a^2 : 2u^2 - (H_{r+s}^{(2)} - H_{r-s}^{(2)}), \quad ac : 4uv - 2(H_{r+s}^{(2)} + H_{r-s}^{(2)}), \\ c^2 : 2v^2 - (H_{r+s}^{(2)} - 2H_s^{(2)} - H_{r-s}^{(2)}). \end{aligned}$$

Theorem 3.2 is equivalent to the vanishing of the last two modulo p . They are *not* rational identities: $\Sigma_{ac} = 0, -26, -\frac{905}{2}, -\frac{167965}{18}, \dots$ and $\Sigma_{c^2} = 0, 13, \frac{905}{4}, \frac{167965}{36}, \dots$, nonzero over $\mathbb{Q}$ for $1 \leq r \leq 30$. Both statements follow from one rational function.

Lemma 4.1 (reflection, rational form). $A_\Gamma(r, z) = \frac{\sin^2 \pi z}{\pi^2} g_r(z)$ with $g_r(z) = \left[\prod_{i=1}^r (z + i) / \prod_{j=0}^r (z - j) \right]^2$, a rational function of degree -2 whose only poles are double poles at $z = 0, 1, \dots, r$.

This replaces the Γ -asymptotics of Theorem 2.3 by an identity. Since $\sin^2 \pi z / \pi^2 = (z - s)^2 - \frac{\pi^2}{3}(z - s)^4 + O((z - s)^6)$ at an integer s , a function $f = \alpha/(z - s)^2 + \beta/(z - s) + \gamma + O(z - s)$ gives $\Phi = (\sin^2 \pi z / \pi^2) f$ regular at s with $\Phi(s) = \alpha$, $\Phi'(s) = \beta$ and $\frac{1}{2}\Phi''(s) = \gamma - \frac{\pi^2}{3}\alpha$. For $f = g_r$ this reads

$$\alpha_s = A(r, s), \quad \beta_s = 2A(r, s)v(r, s), \quad \gamma_s = A(r, s)(2v^2 - (H_{r+s}^{(2)} - 2H_s^{(2)} - H_{r-s}^{(2)})),$$

so $\sum_s \beta_s = 2V_r$ and $\sum_s \gamma_s = \Sigma_{c^2}(r)$; and for $m \geq 1$,

$$g_r(r + m) = C(r, m),$$

which is exactly the ε^2 weight of the borrow digit $s = r + m$. The $\pi^2/3$ of the dictionary cancels the $\zeta(2)$ of $\partial_k^2 \log A_\Gamma$, which is why γ_s is the clean c^2 channel.

Theorem 4.2. $\Sigma_{ac}(r) + 2\Sigma_{c^2}(r) = 0$ for every $r \geq 0$, as an identity over $\mathbb{Q}$.

Proof. $\Sigma_{ac} + 2\Sigma_{c^2} = \sum_{s \leq r} \partial_k [(\partial_n + \partial_k)A_\Gamma](r, s)$; the $\zeta(2)$ terms enter with weights 4 and $2 \cdot (-2)$ and cancel, so this is the unique $\zeta(2)$ -free combination. Now $\sin^2 \pi(z - n)$ is invariant along $(n, z) \mapsto (n + t, z + t)$, and in that direction $(\partial_n + \partial_z) \log g_n = 4[\psi(n + z + 1) - \psi(z + 1)]$ — the $\psi(z - n)$ cancels. Hence at $n = r \in \mathbb{Z}_{\geq 0}$

$$h_r(z) := (\partial_n + \partial_z)g_n(z)|_{n=r} = 4g_r(z) \sum_{i=1}^r \frac{1}{z + i}$$

is rational, with poles only the double poles at $z = 0, \dots, r$ (at $z = -i$ the simple pole meets the double zero of g_r), and $h_r = O(z^{-3})$. A rational function that is $O(z^{-2})$ at infinity has vanishing residue sum, and by the dictionary $\partial_z [(\partial_n + \partial_z)A_\Gamma](r, z)|_{z=s} = \text{Res}_{z=s} h_r$. $\square$

Lemma 4.3. Let $p \geq 5$, $0 \leq r < p$, and let $f \in \mathbb{Q}(z)$ have poles only at $0, \dots, r$, of order ≤ 2 , with p -integral principal parts and $\deg f \leq -2$. Then $\sum_{s=0}^r \text{FP}_{z=s} f + \sum_{t=r+1}^{p-1} f(t) \equiv 0 \pmod{p}$.

Proof. Both sums are sums of $\alpha_{s'}/(y - s')^2 + \beta_{s'}/(y - s')$ over $y \neq s'$ with $0 \leq y \leq p - 1$, all terms p -integral. Interchanging, for fixed s' the difference $y - s'$ runs over all nonzero residues, and $\sum_{w \in \mathbb{F}_p^\times} w^{-1} = \sum_{w \in \mathbb{F}_p^\times} w^{-2} \equiv 0$; the second needs $p \geq 5$. $\square$

Theorem 4.4. For $p \geq 5$ and $0 \leq r < p$, $\Sigma_{c^2}(r) + \Xi_p(r) \equiv 0 \pmod{p}$, where $\Xi_p(r) = \sum_{m=1}^{p-1-r} C(r, m)$.

Proof. Take $f = g_r$ in Lemma 4.3: $\deg g_r = -2$, the principal parts are p -integral ($\alpha_s = A(r, s) \in \mathbb{Z}$, and the only p in a denominator of β_s is the $1/p$ of H_{r+s} when $r + s \geq p$, where $v_p(A(r, s)) = 2$), $\sum_s \gamma_s = \Sigma_{c^2}(r)$, and $\sum_{t=r+1}^{p-1} g_r(t) = \Xi_p(r)$. $\square$

Theorem 3.2 follows, with $X_p(r) = \Sigma_{a^2}(r) + \Xi_p(r)$.

Remark 4.5. The pole set of g_r is $\{0, \dots, r\}$ — the region-I digit range — and its complement in $\mathbb{F}_p$ is $\{r+1, \dots, p-1\}$, the borrow range. So Theorem 4.4 says the finite parts on the pole divisor cancel the values off it, over $\mathbb{F}_p$: it is the residue theorem read over $\mathbb{F}_p$ rather than over $\mathbb{C}$. That is why the vanishing cannot be a rational identity, and indeed cannot be: $\Sigma_{c^2}(r)$ is a fixed nonzero rational for $r \geq 1$ while $\Xi_p(r)$ depends on p . Dropping the borrow region ($\Xi = 0$) breaks Theorem 4.4 at every p and every r with $\Sigma_{c^2}(r) \neq 0$. The contrast with the first order is that there the borrow region contributes nothing at all, because A is a square (Lemma 2.1).

Remark 4.6 ($p = 2, 3$). Both hypotheses $p \geq 5$ are used: Lemma 4.3 needs $\sum_{\mathbb{F}_p^\times} w^{-2} \equiv 0$, and the digit expansion needs $W(M) \equiv ((p-1)!)^M \pmod{p^3}$, i.e. Wolstenholme. At $p = 2, 3$ both fail, as do the regional formulas and Theorem 4.4 at every r ; only Theorem 4.2, being p -free, survives. The conclusion of Theorem 3.2 nevertheless holds at $p = 2$ and $p = 3$ over all 4 and 12 admissible cells. This is a resolution artefact rather than a theorem: the statement is equivalent to $a m_1(\Sigma_{ac} - 2\Xi) + m_2(\Sigma_{c^2} + \Xi) \equiv 0$, whose necessity requires the vectors $(a m_1(a), m_2(a))_{1 \leq a < p}$ to span $\mathbb{F}_p^2$, and their rank is 0 at $p = 2$ and 1 at $p = 3$ against 2 at every $p \geq 5$. With one usable digit, and two, the obstruction cannot be seen.